{3} Current Electricity

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INTRODUCTION

* In the previous two chapters, all charges whether free or bound, were considered to be at rest.

* Charges in motion constitute an electric current.

* In the present chapter, we shall study some of the basic laws concerning steady electric currents.

ELECTRIC CURRENT

* The electric current through any cross-section area is defined as the net charge (ΔQ) flowing through the area per unit time.

$$\boxed{I_{\text{avg.}} = \frac{\Delta Q}{\Delta t} \qquad \text{...... (average current)}}$$

* Instantaneous Current I(t) : The current at time t across the cross-section of the conductor is defined as the value of the ratio of ΔQ to Δt in the limit of Δt tending to zero,

$$\boxed{I(t) = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt} \quad \text{.... instantaneous current}}$$

* SI unit of electric current is ampere (A).

$$\boxed{1\ \text{A} = \frac{1\ \text{C}}{1\ \text{s}}}$$

* The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm^2-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^{+} ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^{6} m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2\ A + (0.6\ A/s^2)\ t^2$.

(a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10$ s?

(b) What constant current would transpot the same charge in same charge in the same time interval?

ELECTRIC CURRENTS IN CONDUCTORS

Conductivity arises due to mobile charge Carreirs

- Metals → Free Electrons
- Ionised gas → electrons and Positive ions
- Electrolyte → Positive and negative ions
- Semiconductors → electrons and holes.

*** In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.**

ELECTRIC CURRENTS IN CONDUCTORS

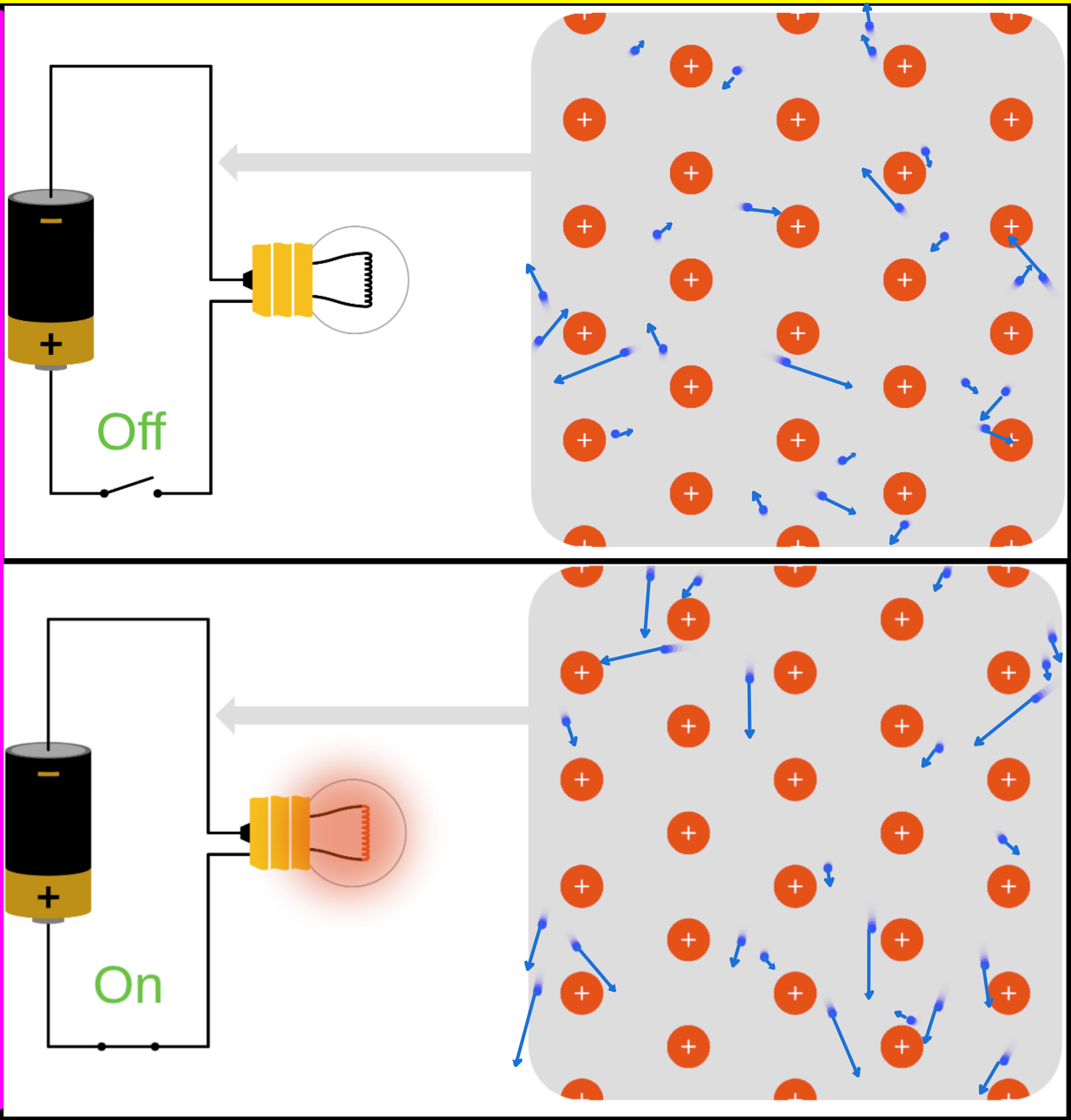
Case 1 : When no elecrtirc filed is present :

* The electrons will be moving due to thermal motion during which they collide with the fixed ions. the direction of its velocity after the collision is completely random.

Thus on the average, the number of electrons travelling in any direction will be equal to the number of electrons travelling in the opposite direction. So, there will be no net electric current.

Case 2 : When elecrtirc filed is present :

* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.



OHM' S LAW

* A basic law regarding flow of currents was discovered by G.S. Ohm in 1828, long before the physical mechanism responsible for flow of currents was discovered.

Imagine a conductor through which a current I is flowing and let V be the potential difference between the ends of the conductor. Then Ohm's law states that the current in a conductor is directly proportional to the voltage applied:

$$V \propto I$$

$$\text{or } \boxed{V = RI}$$

* where the constant of proportionality R is called the resistance of the conductor. The SI units of resistance is ohm (Ω)

* The resistance R not only depends on the material of the conductor but also on the dimensions (length l and cross sectional area A) of the conductor.

* In general, then resistance is proportional to length, $R \propto l$

* The resistance R is inversely proportional to the cross-sectional area, $R \propto 1/A$

$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

* where the constant of proportionality called resistivity (ρ) depends on the material

CURRENT DENSITY AND EQUIVALENT FORM OF OHM'S LAW

* **Current Density (J) :** Current per unit area (taken normal to the current), I/A, is called current density and is denoted by J. It is a vector quantity. The SI units of the current density are A/m^2.

$$\boxed{dI = \vec{J}\cdot d\vec{A} = J\,dA\cos\theta}$$

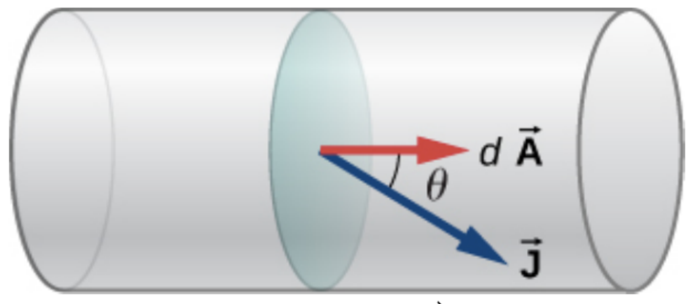


Figure 9.12 The current density $\vec{\mathbf{J}}$ is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.

* **Equivalent form of Ohm's Law :**

If E is the magnitude of uniform electric field in the conductor whose length is l, then the potential difference V across its ends is V = El

$$\text{From Ohm's Law} : V = IR$$

$$El = JAR \qquad \left(\because V = El \text{ and } J = \frac{I}{A}\right)$$

$$E = \frac{RA}{l}J \qquad \text{or} \qquad \boxed{E = \rho J} \quad \left(\because \rho = \frac{RA}{l}\right)$$

3.6 A negligibly small current is passed through a wire of length 15 m and uniform cross-section 6.0×10^{-7} m^2, and its resistance is measured to be 5.0 Ω. What is the resistivity of the material at the temperature of the experiment?

Ex 6 : A wire 50 cm long and 1 mm^2 in cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is

$a)\ 2 \times 10^{-7}\ \Omega m$ $\qquad$ $b)\ 5 \times 10^{-7}\ \Omega m$

$c)\ 4 \times 10^{-6}\ \Omega m$ $\qquad$ $d)\ 1 \times 10^{-6}\ \Omega m$

Ex 7: A wire of length L and cross-section area A has resistance R. What will be the resistance of the wire if it is stretched to twice its original length? Assume that the density and resistivity of material do not change when the wire is stretched.

Ex 8 : A wire of length L is drawn such that its diameter is reduced to half of its original diameter. If the initial resistance of the wire is 10 Ω, its new resistance would be

a) 40 Ω (b) 80 Ω (c) 120 Ω (d) 160 Ω

DRIFT VELCOITY

* In metals the charge carriers are negatively charged electrons that move to create an electric current.

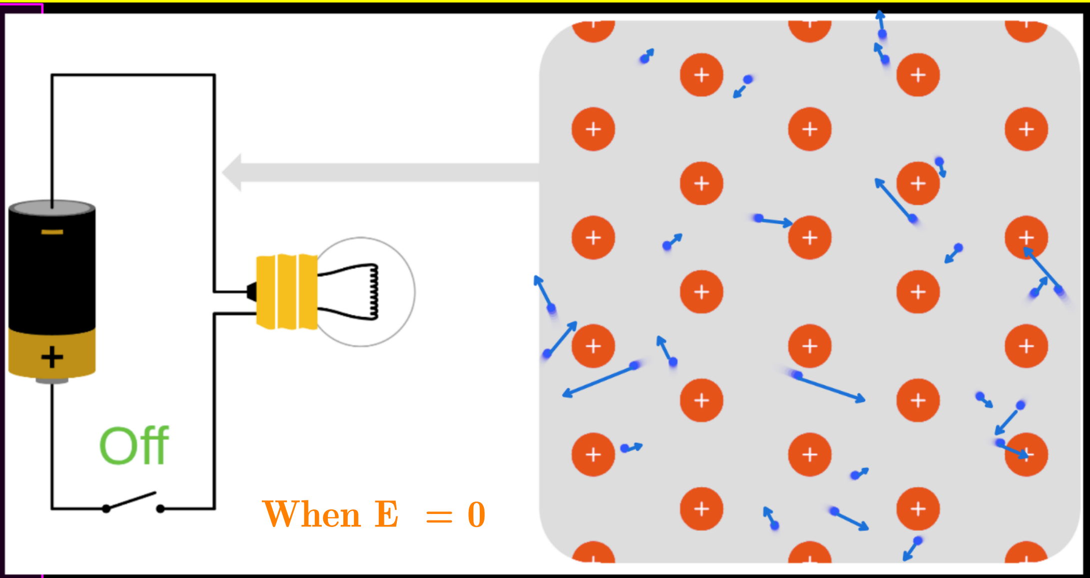
* Without an applied electric field, electrons move around randomly in a material.

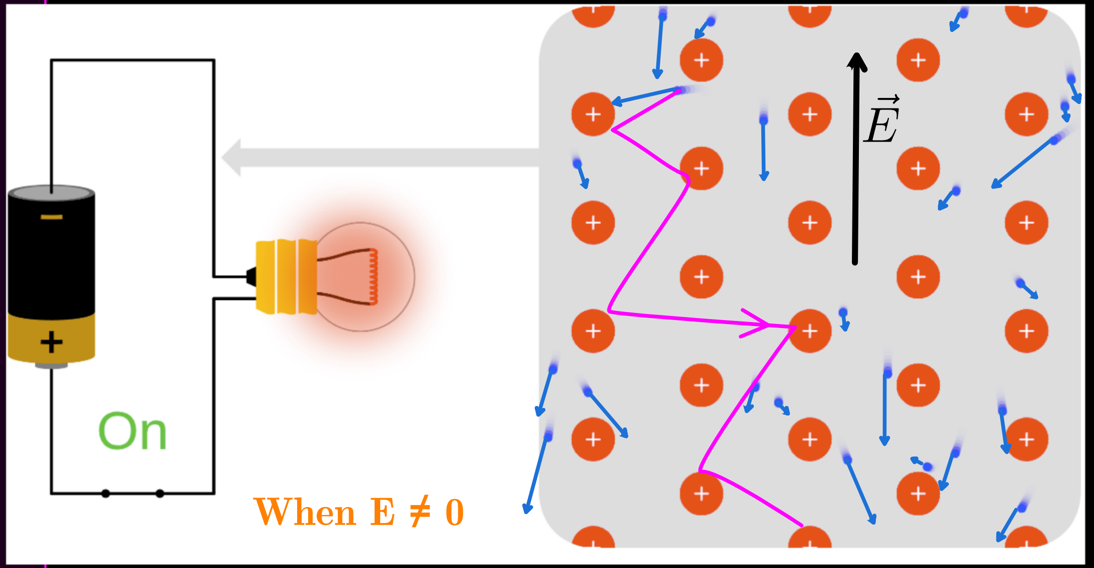
* If we consider all the n electrons with random velocities (u_1 , u_2, u_3, ... , u_n), their average velocity will be zero since their directions are random.

$$\vec{u}_{\text{avg.}} = \frac{\vec{u}_1 + \vec{u}_2 + ... + \vec{u}_n}{n} = \frac{1}{n}\sum_{i=0}^{n} u_i = 0$$

* When an electric field is present. Electrons will experience a force in the oppsite direction of the Electric filed.

$$\vec{F} = -e\vec{E} \qquad \therefore \text{ acceleration } \boxed{\vec{a} = \frac{-e\vec{E}}{m_e}}$$





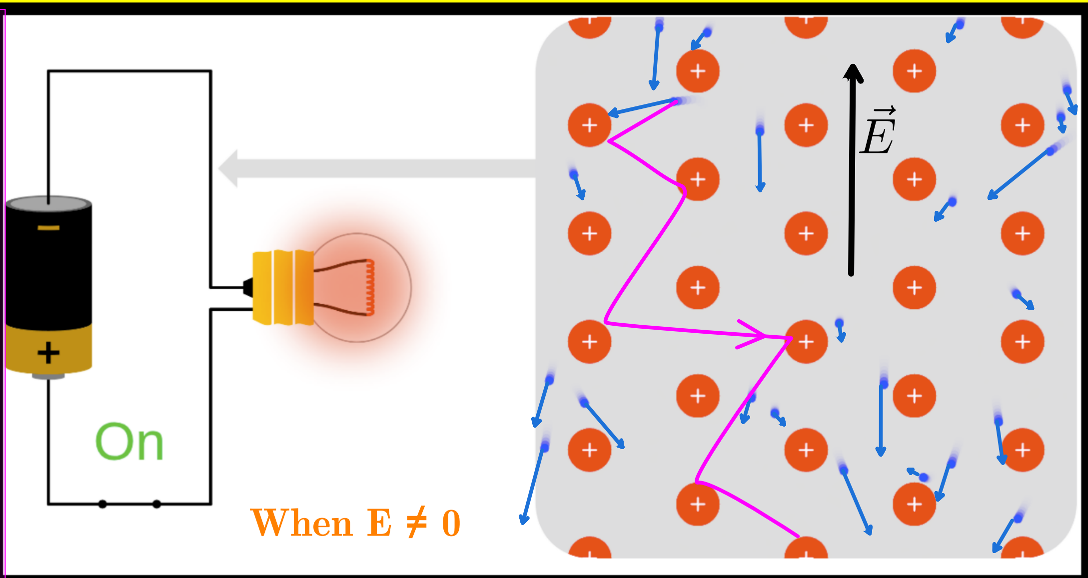
DRIFT VELCOITY

* As the electron accelerates it gains velocity, which last for a short time and is lost in the next collision.

* The velocity gained by ith electron between two successive collisions.

 (t_i - time between two successice collsion)

$$\vec{v}_i = \vec{u}_i + \vec{a}\, t_i \qquad \vec{v}_i = \vec{u}_i - \frac{e\vec{E}}{m_e}\, t_i$$



* The average velocity gained by all n elecrtrons (called drift velocity) will be :

$$\vec{v}_{\text{avg.}} = \frac{1}{n}\sum_{i=1}^{n} v_i = \frac{1}{n}\sum_{i=1}^{n} u_i - \frac{e\vec{E}}{m_e}\sum_{i=1}^{n}\frac{t_i}{n}$$

$$\vec{v}_d = 0 - \frac{e\vec{E}}{m_e}\tau$$

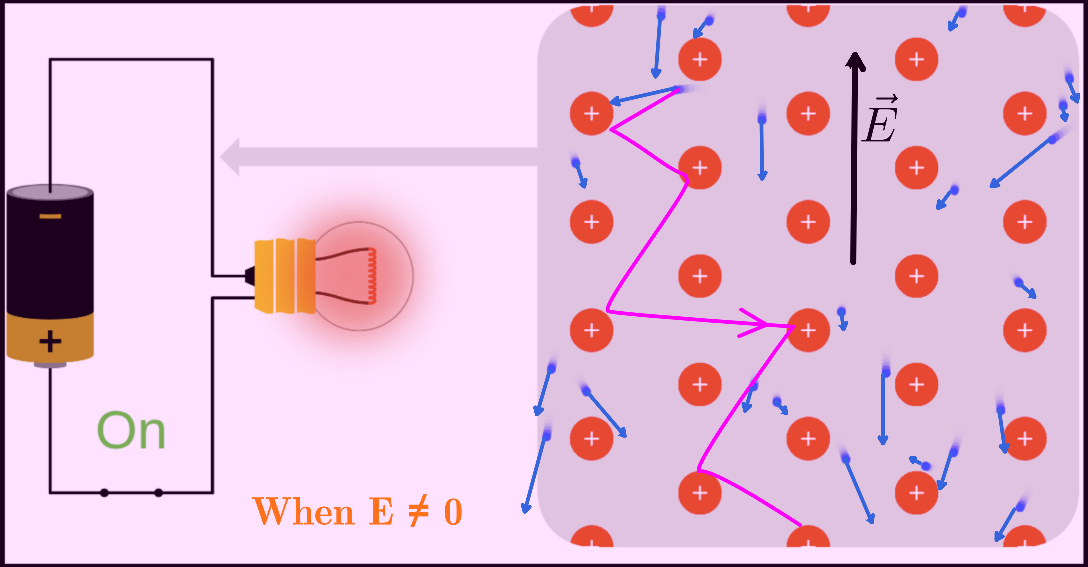
* τ is the average time between two successive collission and is known as relaxation time.

$$\boxed{\vec{v}_d = -\frac{e\vec{E}}{m_e}\tau}$$

* Drift velocity : The average velocity with which the free electrons get drifted in the opposite direction of the applied electric field.

DRIFT VELCOITY

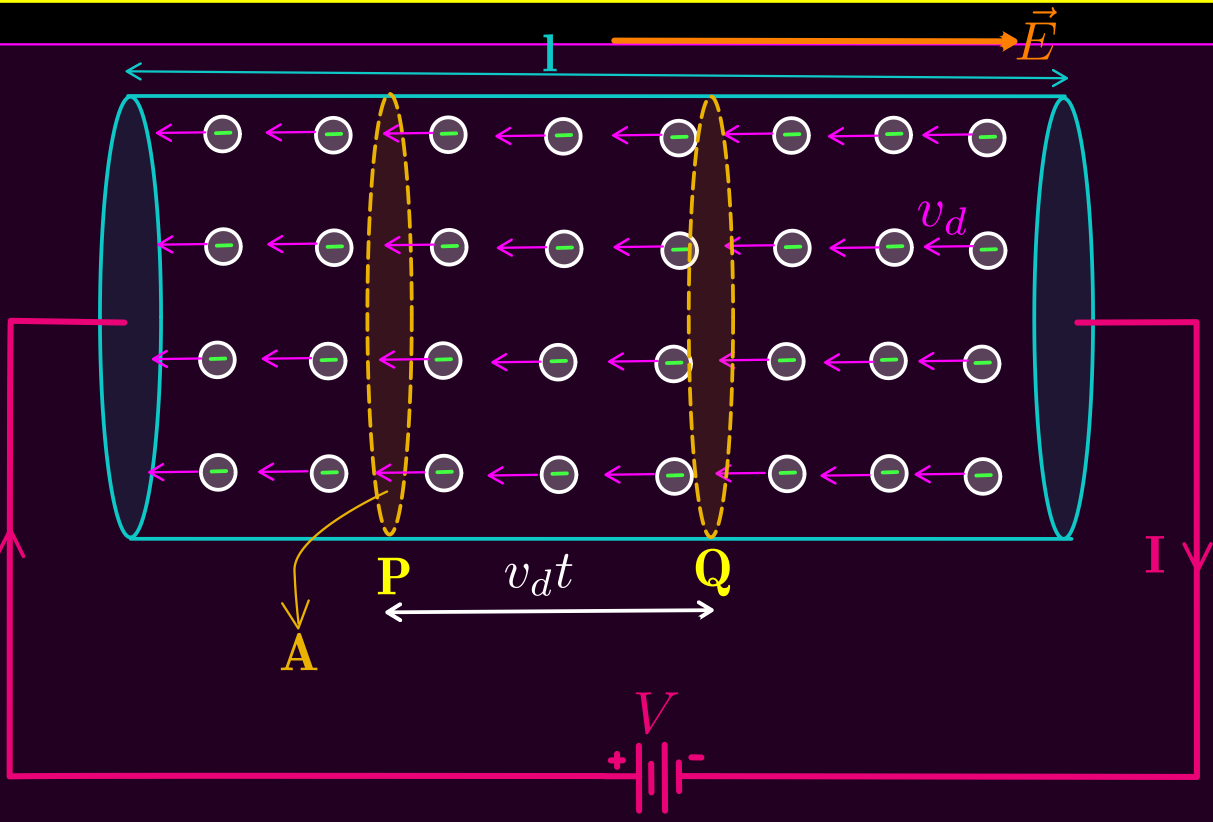
* Due to continuous collisions of electrons and the positive ions, the speed of the electrons doesn't continuously increase. The result of this is that the electrons move within the material with some average constant velocity (drift velocity) in the direction opposite the applied electric field.



* OR the drift velocity is independent of time.

* The drift velocity of electrons is very small in the order of 10^{-4} ms^{-1}.

RELATION BETWEEN CUREENT AND DRIFT VELOCITY



* Consider a conductor of length l and uniform cross-sectional area A.

* A constant potential difference V is maintained across its ends. Hence an electric filed E = Vl is set up in the wire.

* Under the influence of this electric field, the electrons drift with a constant velocity v_d in the opposite direction to that of E

* In a small time t, all the elctrons upto the distance $v_d t$ will cross the plane area P.

* Let, The number of free charges per unit volume, or the number density of free charges, is given the symbol n where n = Number of charges/Volume

* Therfore, number of electron crossing the plane area P in time t $= n \times$ Volume of cylinder PQ

$$= n \times A v_d \; t$$

RELATION BETWEEN CUREENT AND DRIFT VELOCITY

* **Total charge (q) flowing through the cross-section area (A) in time t is**

$$q = \text{no. of electrons} \times e$$

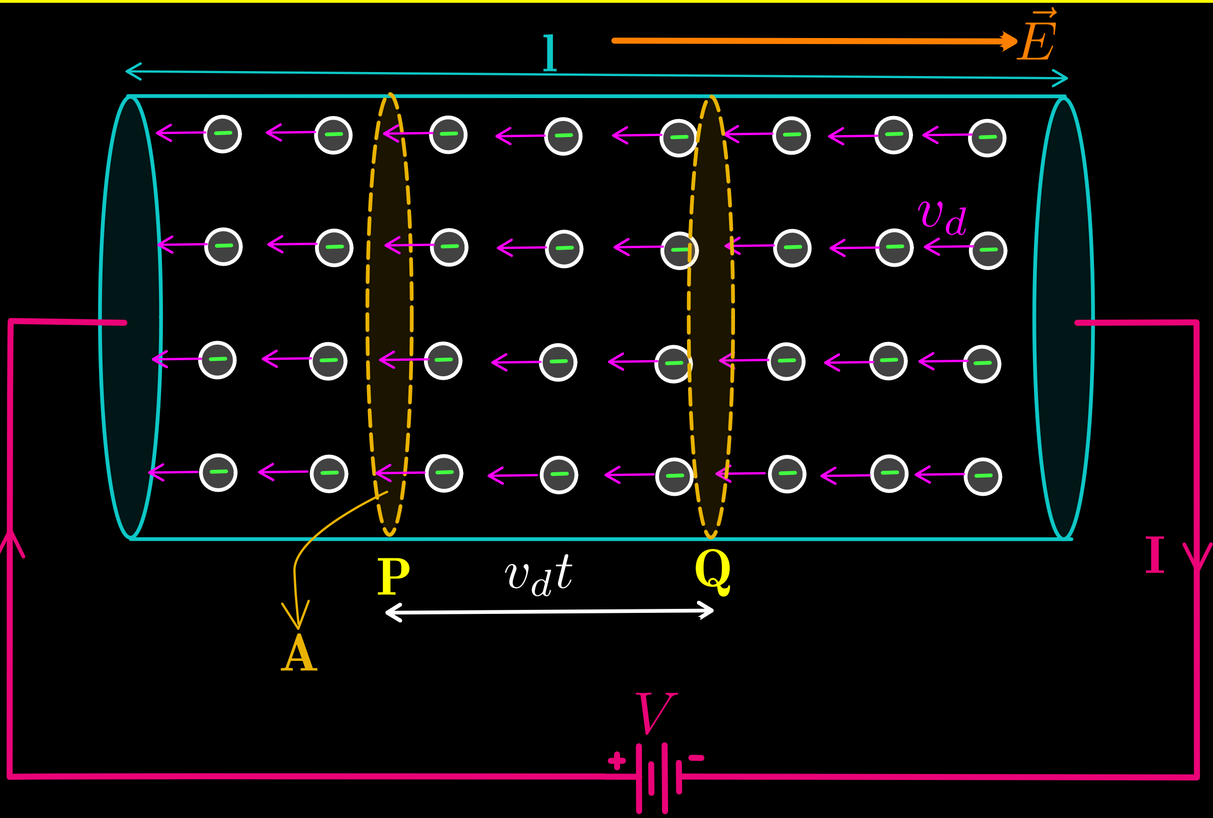
$$q = nAv_d\, t \times e$$

* **Hence, the current I in the wire is given by:**

$$I = \frac{q}{t}$$

$$I = \frac{neAv_dt}{t}$$

$$\boxed{I = neAv_d}$$



ORIGIN OF RESISTIVITY

* We have already derived that :

$$I = neAv_d \qquad \text{and} \quad v_d = \frac{eE\tau}{m_e}$$

$$I = neA \times \frac{eE\tau}{m_e}$$

$$\frac{I}{A} = \frac{ne^2\tau}{m_e}E$$

$$J = \frac{ne^2\tau}{m_e}E \quad \left(\because J = \frac{I}{A}\right)$$

Also, $J = \sigma E$ (Ohm's law)

$$\therefore \boxed{\text{conductivity } \sigma = \frac{ne^2\tau}{m_e}}$$

and $\boxed{\text{Resistivity } \rho = \frac{1}{\sigma} = \frac{m_e}{ne^2\tau}}$

MOBILITY

* **The mobility μ is defined as the magnitude of the drift velocity per unit electric field:**

$$\mu = \frac{v_d}{E}$$

$$\text{Since, } v_d = \frac{eE\tau}{m}$$

$$\boxed{\therefore \mu = \frac{v_d}{E} = \frac{e\tau}{m}}$$

Ex 9: There is a current of 40 A in a wire of 10^{-6} square metre area of cross-section, if the number of free electrons per cubic metre is 10^{29}, then the drift velcoity will be

$(a)\ 1.25 \times 10^{3}\ m/s$ $\quad$ $(b)\ 2.5 \times 10^{-3}\ m/s$

$(c)\ 25 \times 10^{-3}\ m/s$ $\quad$ $(d)\ 25 \times 10^{3}\ m/s$

Example 3.1 (a) Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area 1.0×10^{-7} m^2 carrying a current of 1.5 A. Assume that each copper atom contributes roughly one conduction electron. The density of copper is 9.0×10^3 kg/m^3, and its atomic mass is 63.5 u. (b) Compare the drift speed obtained above with, (i) thermal speeds of copper atoms at ordinary temperatures, (ii) speed of propagation of electric field along the conductor which causes the drift motion.